Open Critical Mapping Student Workshop

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Packages

Let's start by installing/loading packages (as needed) and libraries:

```
# Load necessary libraries
library(tidycensus)
library(tidyverse)
#library(dplyr) if tidyverse is not working
#library(ggplot2) if tidyverse is not working
library(sf)
library(viridis)
library(mapview)
```

U.S. Census API key

To access the census data for mapping, please sign up for an API key using this link. You should receive an API key in your email. Place it in the code chunk as indicated below.

#Var Details Variable 2: Commuter Vehicle Reliance Numerator (Variable): B08301_002E (Estimate!!Total:!!Car, truck, or van) Denominator (Universe): B08301_001E (Total workers 16 years and over) Table: B08301 (Means of Transportation to Work) Field Calculation: (B08301_002E / B08301_001E) * 100 % Workers Commuting by Private Vehicle

Introducing Your Variable

Fulton 2017 Setup

```
fulton_cvr1 <- get_acs(
  geography = "tract",
  variables = "B08301_002E",
  state = "GA",
  county = "Fulton",
  year = 2017,
  geometry = TRUE
)

fulton_cvr1
```

Fulton 2024 Setup

```
fulton_cvr2 <- get_acs(
  geography = "tract",
  variables = "B08301_002E",
  state = "GA",
  county = "Fulton",
  year = 2024,
  geometry = TRUE
)

fulton_cvr2
```

Georgia 2017 Setup

```
georgia_cvr1 <- get_acs(
  geography = "tract",
  variables = "B08301_002E",
  state = "GA",
  #county = "Fulton",
  year = 2017,
  geometry = TRUE
)

georgia_cvr1
```

Georgia 2024 Setup

```
georgia_cvr2 <- get_acs(
  geography = "tract",
  variables = "B08301_002E",
  state = "GA",
  #county = "Fulton",
  year = 2024,
  geometry = TRUE
)

georgia_cvr2
```

Fulton Black Dot Setup

```
fulton_blackpop <- get_acs(
  geography = "tract",
  variables = "B01001B_001", # Black population variable
  state = "GA",
  county = "Fulton",
  year = 2017,
  geometry = T
)
```

Black Dot Setup

```
blackpop_dots <- as_dot_density(
  fulton_blackpop,
  value = "estimate",
  values_per_dot = 1000 # adjust this number to
)
```

Durham Setup

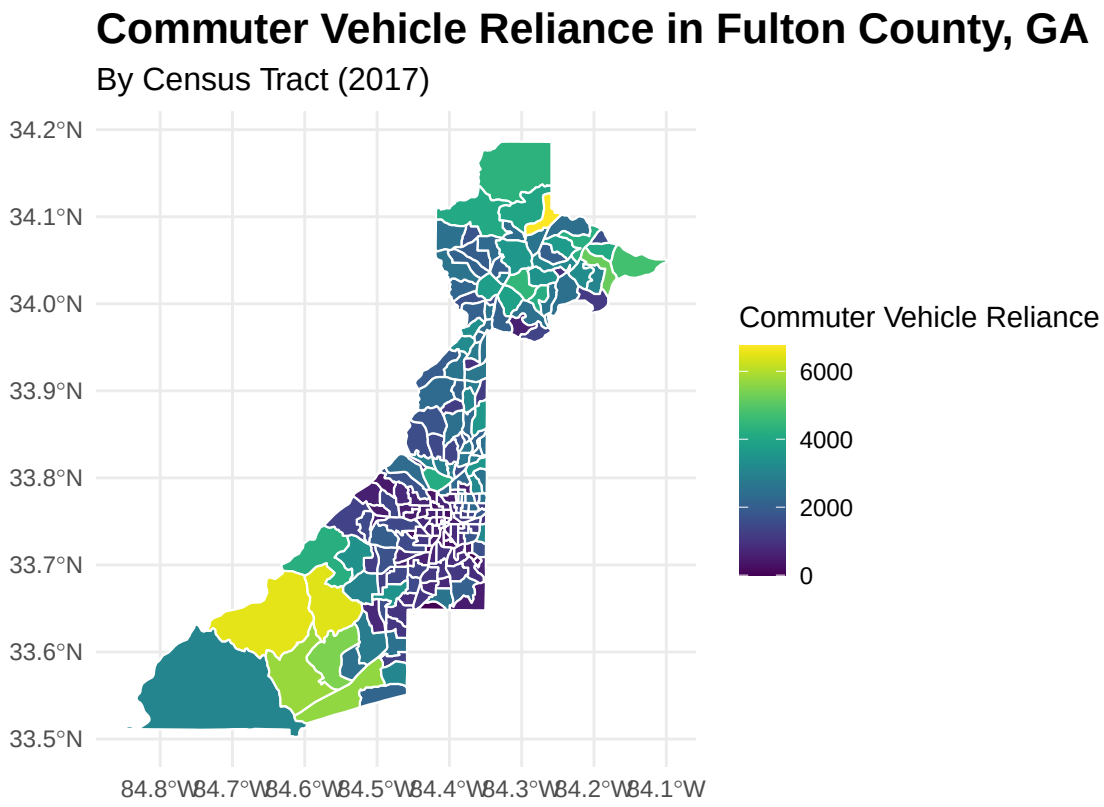
```
durham_cvr <- get_acs(
  geography = "tract",
  variables = "B08301_002E",
  state = "NC",
  county = "Durham",
  year = 2023,
  geometry = TRUE
)

durham_cvr
```

Mapping

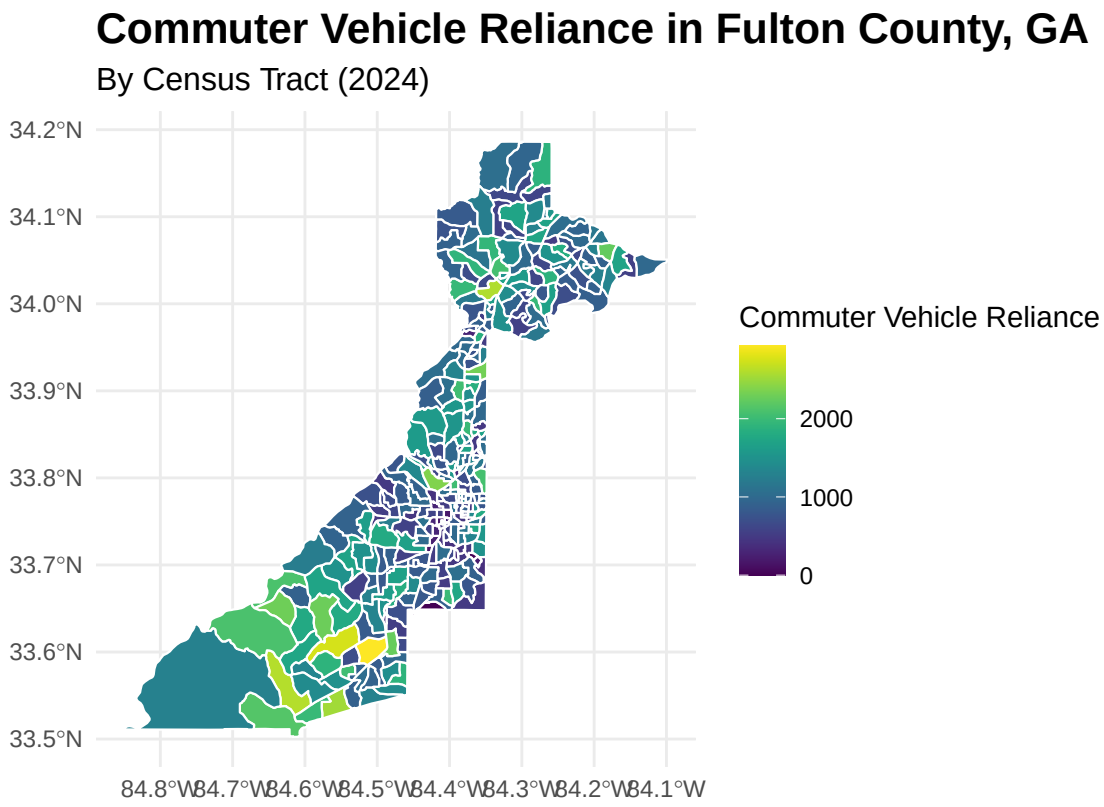
Fulton CVR 2017

```
ggplot(fulton_cvr1) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.4) +
  scale_fill_viridis(option = "blue", name = "Commuter Vehicle Reliance") +
  labs(
    title = "Commuter Vehicle Reliance in Fulton County, GA",
    subtitle = "By Census Tract (2017)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    plot.caption = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



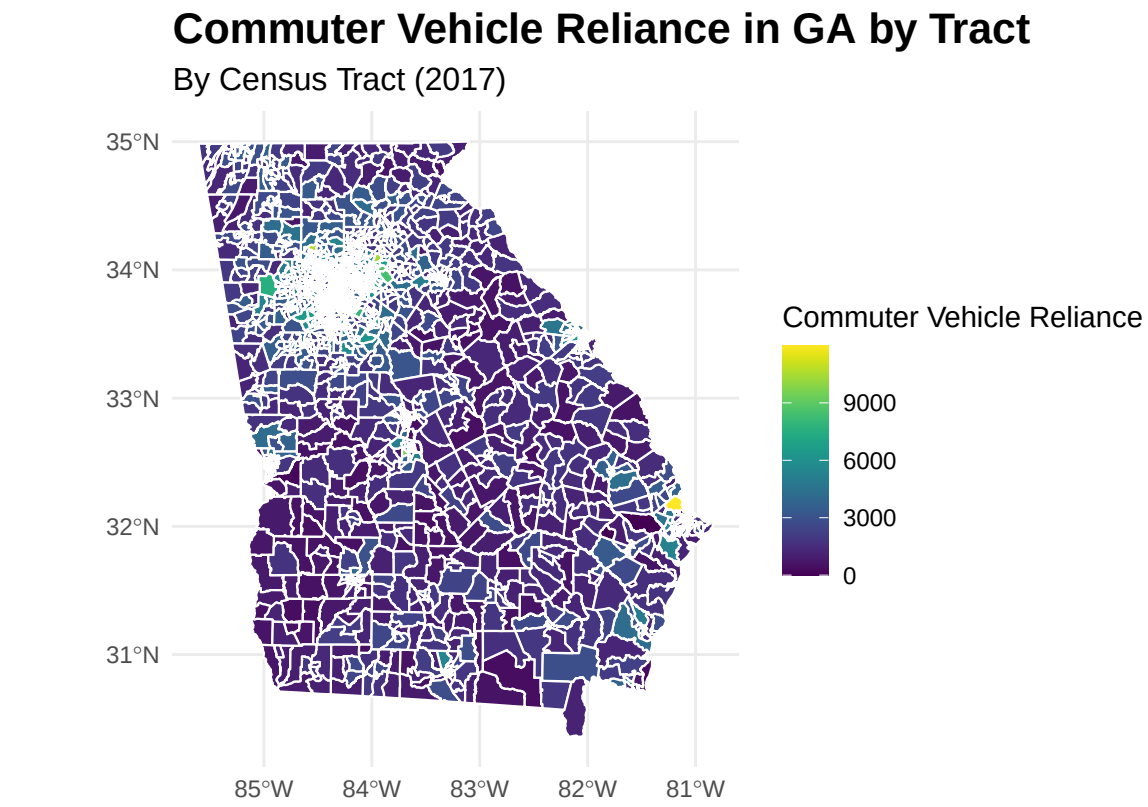
Fulton CVR 2024

```
ggplot(fulton_cvr2) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.4) +
  scale_fill_viridis(option = "blue", name = "Commuter Vehicle Reliance") +
  labs(
    title = "Commuter Vehicle Reliance in Fulton County, GA",
    subtitle = "By Census Tract (2024)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    plot.caption = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



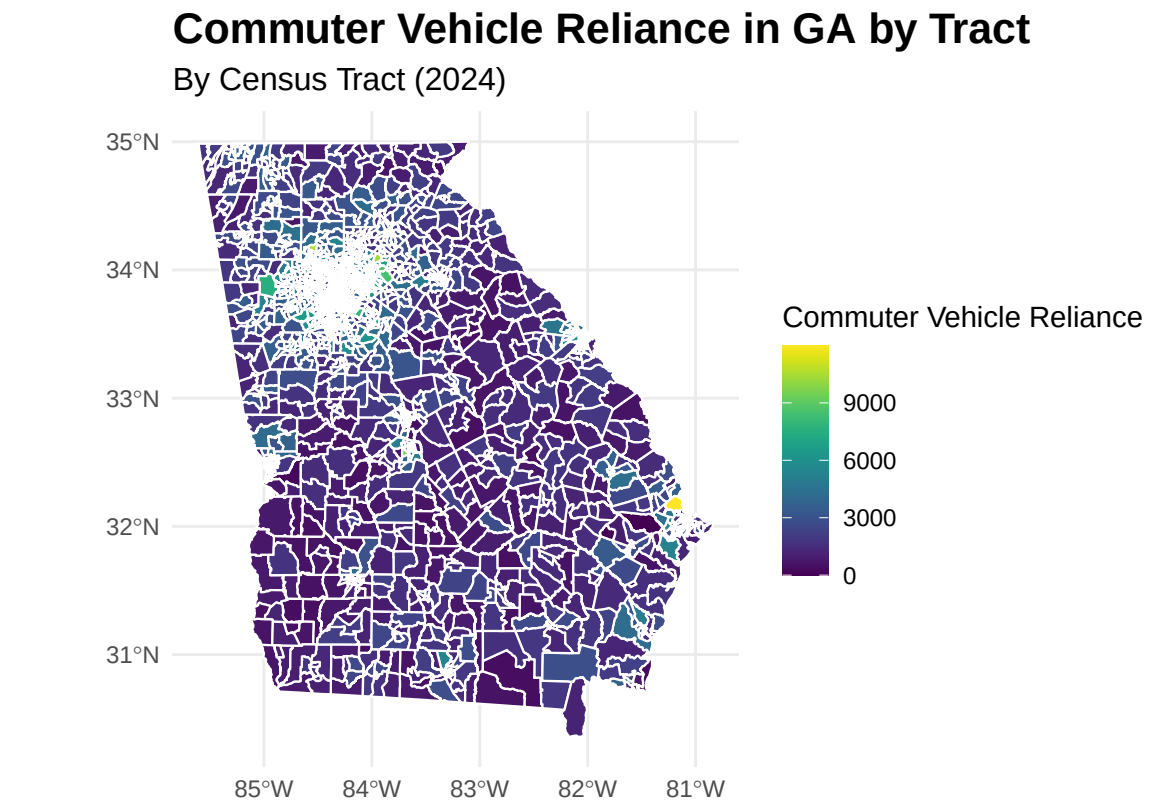
Georgia CVR 2017

```
ggplot(georgia_cvr1) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.4) +
  scale_fill_viridis(option = "blue", name = "Commuter Vehicle Reliance") +
  labs(
    title = "Commuter Vehicle Reliance in GA by Tract",
    subtitle = "By Census Tract (2017)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    plot.caption = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



Georgia CVR 2024

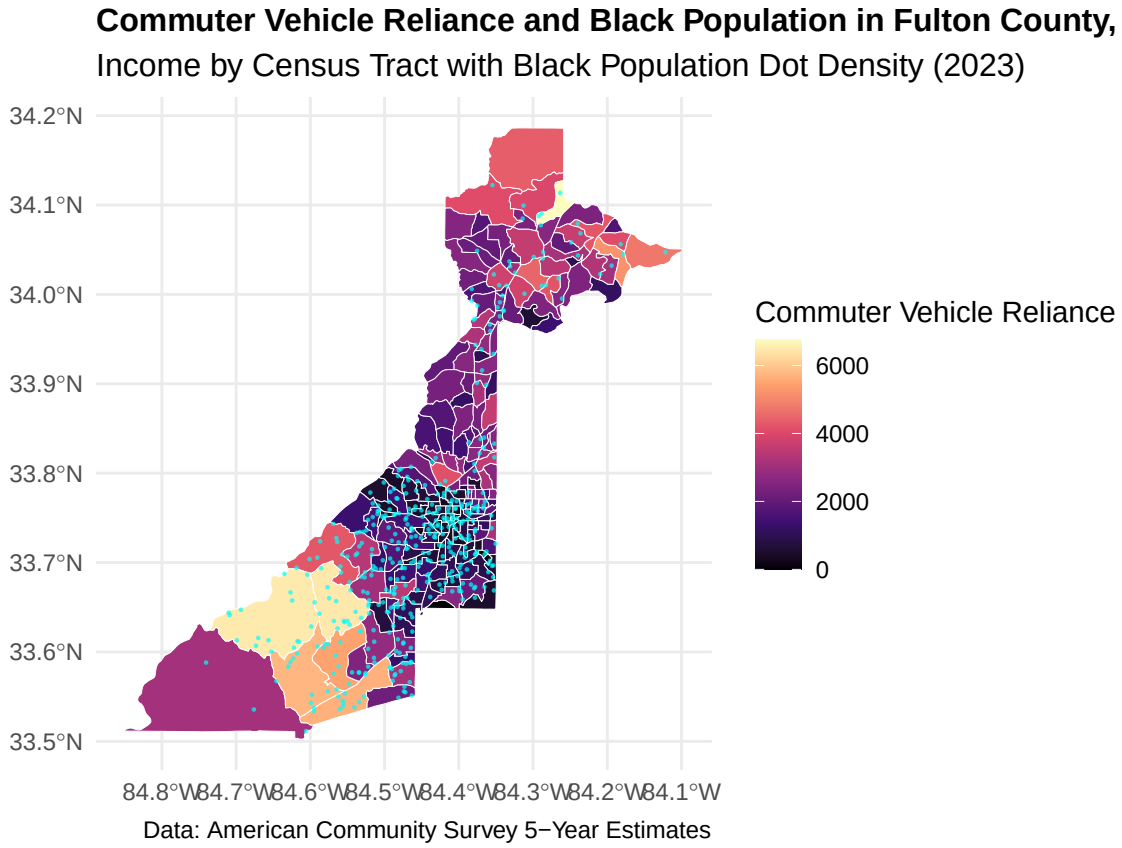
```
ggplot(georgia_cvr1) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.4) +
  scale_fill_viridis(option = "blue", name = "Commuter Vehicle Reliance") +
  labs(
    title = "Commuter Vehicle Reliance in GA by Tract",
    subtitle = "By Census Tract (2024)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    plot.caption = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



Fulton 2017 CVR with Black Dot

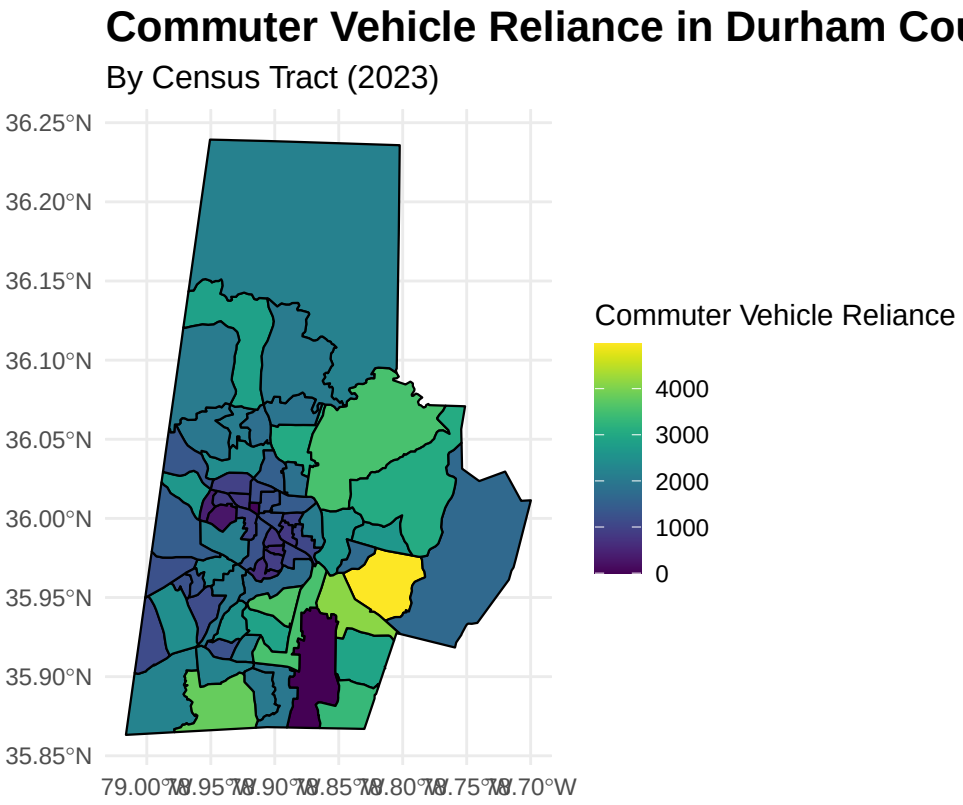
```
ggplot() +
  geom_sf(
    data = fulton_cvr1,
    aes(fill = estimate),
    color = "white",
    linewidth = 0.1
  ) +
  geom_sf(
    data = blackpop_dots,
    color = "cyan",
    size = 0.15,
    alpha = 0.7,
    inherit.aes = FALSE
  ) +
  scale_fill_viridis_c(
    option = "magma",
    name = "Commuter Vehicle Reliance"
  ) +
  labs(
    title = "Commuter Vehicle Reliance and Black Population in Fulton County, GA",
    subtitle = "Income by Census Tract with Black Population Dot Density (2023)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
```

```
  theme(
    plot.title = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



Durham CVR

```
ggplot(durham_cvr) +
  geom_sf(aes(fill = estimate), color = "black", size = 0.4) +
  scale_fill_viridis(option = "yellow", name = "Commuter Vehicle Reliance") +
  labs(
    title = "Commuter Vehicle Reliance in Durham County, NC",
    subtitle = "By Census Tract (2023)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    plot.caption = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```



: American Community Survey 5−Year Estimates